

Dr Sam Harrison

Contact Address: UK Centre for Ecology & Hydrology, Lancaster, LA1 4AP
Job Title: Environmental Modeller
Website: samharrison.science

Professional and Educational Qualifications

PhD (Quantum Nanotechnology)	Department of Physics, Lancaster University	2016
MPhys (Physics)	Department of Physics, Lancaster University	2012

Summary of Professional Expertise

Dr Harrison is an environmental modeller specialising in the emissions, fate, exposure and risk of chemicals and materials in the environment, with a focus on emerging contaminants including plastics, pharmaceuticals, nanomaterials and PFAS. His work increasingly sits at the science–policy interface for chemicals, waste and pollution. He leads UKCEH's engagement in the negotiations towards a Global Plastics Treaty, and is UKCEH Head of Delegation to the first International Conference of the Global Framework on Chemicals (IC1, Geneva, November 2026). He has provided evidence to UK government, including prioritising chemical exposure pathways, assessing the environmental and economic impacts of risk mitigation options for intentionally added microplastics, and reviewing stakeholder needs in chemical exposure modelling.

He leads and participates in international consortia spanning academia, industry and regulators, developing models of plastic emissions, fragmentation and multimedia fate, and assessing the environmental risks of pharmaceuticals and nanomaterials through Horizon Europe, NERC and Cefic-funded projects. A central thread of his work is making models relevant and usable for regulators and decision-makers: he develops open-source, FAIR modelling tools with graphical user interfaces and data visualisations, and has led international efforts towards community-based, open environmental modelling. He chairs a long-standing session on advances in exposure modelling at the Society of Environmental Toxicology and Chemistry's Europe annual meeting and is a Fellow of the Institute for Research Software (formerly Software Sustainability Institute).

Employment History

2017 - Present Environmental Modeller at UK Centre for Ecology & Hydrology, Lancaster

Selected Research Projects and Funding

2026 - 2030	BIO-EMISSOME: Lifecycle environmental emissions and impacts of bio-based materials. <i>Institutional PI and task lead. Horizon Europe.</i>
2025 - 2028	P-FASE: Developing spatiotemporal model of PFAS environmental exposure in UK. <i>NERC.</i>
2025 - 2027	FRAGMENT-ARTICLE: Degradation and fragmentation of plastic articles. <i>European Chemical Industry Council (Cefic).</i>
2025 - 2026	Environmental Data Futures. <i>Co-PI of project exploring long-term implications of societal and technological trends shaping data systems. Internal funding.</i>
2025	Microplastics model integration: <i>Institutional PI of project integrating models of microplastic emissions and exposure. Cefic.</i>
2024 - 2026	Biodegradable polymers: <i>PI of project to model the degradation and fragmentation of biodegradable polymers. BASF SE.</i>
2023 - 2024	Cost appraisal of mitigation options against intentionally added microplastics. <i>Defra.</i>

2023 - 2024	EMIFACT-MNP: EMission FACTors of Micro and NanoPlastics. <i>PI</i> of international consortium (Europe and US) to develop a model of plastic releases to the environment. <i>Cefic</i> .
2022 - 2024	DEPICTION: DEveloPing an International CollaboraTion to advance community-based, Open and FAIR eNvironmental modelling. <i>PI</i> of international consortium. <i>NERC</i> .
2022 - 2026	ETERNAL: Boosting thE reducTion of the EnviRonmeNtal impact of phArmaceutical products throughout their entire life cycle. <i>Institutional PI and WP lead. Horizon Europe</i> .
2021 - 2023	FRAGMENT-MNP: Developing a mechanistic model of Micro and NanoPlastic FRAGMentation in the ENviroNmenT. <i>PI</i> of international consortium. <i>Cefic</i> .
2021 - 2023	UTOPIA: Development of a mUltimedia uniT world OPen-source model for mIcroplAstic. <i>Institutional PI</i> , coordinator Stockholm University. <i>Cefic</i> .
2018 - 2023	UKSCaPE SPEED. Development of a model for long-term spatiotemporal prediction of metal concentrations in topsoils for the UK, including future projections based on potential climatic and socioeconomic pathways. <i>NERC</i> .
2018	Stakeholder needs in environmental chemical exposure modelling. <i>Defra</i> .
2017 - 2019	NanoFASE: Nanomaterial FAte and Speciation in the Environment. Lead developer of a spatiotemporal model for prediction of engineered nanomaterial fate in soils, surface waters and sediments. <i>H2020</i> .

Memberships: Society of Environmental Toxicology and Chemistry (SETAC), Society of Research Software Engineering (SocRSE), Community Surface Dynamics Modeling System (CSDMS).

Selected Research Outputs

- S. Shamsudheen, **S. Harrison** et al. *Global plastic waste exports and the implications for continued microplastic pollution: toward a science policy interface for sustainable futures*. npj Emerging Contaminants 2 (2026). DOI: 10.1038/s44454-026-00047-z
- C. Uluseker ... **S. Harrison**. *Beyond source restrictions: Combined mitigation strategies substantially reduce emissions and exposure of intentionally added microplastics in the United Kingdom*. Integrated Environmental Assessment and Management (2026). DOI: 10.1093/inteam/vjag103.
- C. Uluseker ... **S. Harrison**. *Advances and challenges in modelling the environmental fate and exposure of pharmaceuticals: a comprehensive review*. Environmental Science: Processes & Impacts 27 (2025). DOI: 10.1039/d5em00449g
- S. Harrison** et al. *Continuous Improvement towards Environmental Protection for Pharmaceuticals: Advancing a Strategy for Europe*. Environmental Sciences Europe 37 (2025). DOI: 10.1186/s12302-025-01180-z.
- S. Harrison** et al. *FRAGMENT-MNP: A model of micro- and nanoplastic fragmentation in the environment*. Journal of Open Source Software 10 (2025). DOI: 10.21105/joss.08061.
- E. Dearlove, **S. Harrison** et al. *Agrochemical inputs to managed oil palm plantations are a probable risk to ecosystems: Results from a screening level risk assessment*. Environmental Pollution 361 (2024). DOI: 10.1016/j.envpol.2024.124749.
- M. MacLeod ... **S. Harrison** et al. *Computational models to confront the complex pollution footprint of plastic in the environment*. Nature Computational Science 3 486 (2023). DOI: 10.1038/s43588-023-00445-y.
- S. Harrison**. *Predicting the fate of plastic in the environment*. Nature Water 1 490 (2023). DOI: 10.1038/s44221-023-00097-2.
- C. Svendsen ... **S. Harrison** et al. *Key principles and operational practices for improved nanotechnology environmental exposure assessment*. Nature Nanotechnology 15 731 (2020). DOI: 10.1038/s41565-020-0742-1
- R. Williams, **S. Harrison** et al. *Models for assessing engineered nanomaterial fate and behaviour in the aquatic environment*. Current Opinion in Environmental Sustainability 36 105 (2019). DOI: 10.1016/j.cosust.2018.11.002.